

1. Define and populate Rate and Customer tables in two separate databases. Suggested that the Rate table be in a Rate database and the Customer table be in a Customer database.

```
SQLQuery14.s...Q\praty (79))*  SQLQuery12.s...Q\praty (78))* ✕
1  -- Create Rate table
2
3  CREATE TABLE Rate
4  (
5      RateID INT IDENTITY(1,1) NOT NULL,
6      Gender CHAR(1) NOT NULL,
7      Rate FLOAT NOT NULL,
8      CONSTRAINT PK_Rate PRIMARY KEY (RateID)
9  );
```

```
q\SQLEXPRES...te - dbo.Rate ✕  SQLQuery14....\praty (
1  -- Populating Rate Table
2
3  INSERT INTO Rate (Gender, Rate)
4  VALUES
5  ('M', 1.00), ('F', 0.95),
6  ('M', 1.05), ('F', 1.00),
7  ('M', 1.10), ('F', 1.05),
8  ('M', 1.15), ('F', 1.10),
9  ('M', 1.20), ('F', 1.15),
10 ('M', 1.25), ('F', 1.20),
11 ('M', 1.30), ('F', 1.25),
12 ('M', 1.35), ('F', 1.30),
13 ('M', 1.40), ('F', 1.35),
14 ('M', 1.50), ('F', 1.45),
15 ('M', 1.60), ('F', 1.55),
16 ('M', 1.70), ('F', 1.65),
17 ('M', 1.80), ('F', 1.75),
18 ('M', 1.95), ('F', 1.85),
19 ('M', 2.10), ('F', 2.00),
20 ('M', 2.30), ('F', 2.15),
21 ('M', 2.50), ('F', 2.35),
22 ('M', 2.75), ('F', 2.55),
23 ('M', 3.00), ('F', 2.80),
24 ('M', 3.30), ('F', 3.10);
```

```
SQLQuery15....\praty (73))*  loq\SQLEXP...te - dbo.Rate  SQLQuery14..
1  -- Create Customer Table
2
3  CREATE TABLE Customer
4  (
5      CustomerID INT IDENTITY(1,1) NOT NULL,
6      FName NVARCHAR(50) NOT NULL,
7      LName NVARCHAR(50) NOT NULL,
8      DOB DATE NOT NULL,
9      Gender CHAR(1) NOT NULL, -- M = Male | F = Female
10     CONSTRAINT PK_Customer PRIMARY KEY (CustomerID)
11 );
```

```
loq\SQLEXP...dbo.Customer  SQLQuery16....\praty (88))*  SQLQuery15.s...Q\praty (73))*  loq\SQLEXP
1  -- Populating Customer Table
2
3  INSERT INTO Customer (FName, LName, DOB, Gender)
4  VALUES
5  ('Aiden', 'Smith', '2026-02-10', 'M'), ('Liam', 'Brown', '2022-06-14', 'M'),
6  ('Noah', 'Johnson', '2018-09-01', 'M'), ('Emma', 'Davis', '2015-11-20', 'F'),
7  ('Olivia', 'Miller', '2012-04-18', 'F'), ('Sophia', 'Wilson', '2009-01-10', 'F'),
8  ('James', 'Moore', '2006-07-22', 'M'), ('Logan', 'Taylor', '2003-02-14', 'M'),
9  ('Mia', 'Anderson', '2000-09-30', 'F'), ('Ethan', 'Thomas', '1998-12-01', 'M'),
10 ('Isabella', 'Jackson', '1995-05-16', 'F'), ('Lucas', 'White', '1992-08-09', 'M'),
11 ('Amelia', 'Harris', '1989-03-03', 'F'), ('Henry', 'Martin', '1986-10-11', 'M'),
12 ('Charlotte', 'Thompson', '1983-06-25', 'F'), ('Benjamin', 'Garcia', '1980-01-19', 'M'),
13 ('Evelyn', 'Martinez', '1978-04-14', 'F'), ('Alexander', 'Robinson', '1975-09-09', 'M'),
14 ('Harper', 'Clark', '1972-12-22', 'F'), ('Daniel', 'Rodriguez', '1970-02-07', 'M'),
15 ('Ella', 'Lewis', '1968-07-30', 'F'), ('Matthew', 'Lee', '1965-11-18', 'M'),
16 ('Avery', 'Walker', '1962-03-25', 'F'), ('Joseph', 'Hall', '1960-06-06', 'M'),
17 ('Scarlett', 'Allen', '1958-09-14', 'F'), ('David', 'Young', '1955-01-08', 'M'),
18 ('Grace', 'King', '1952-05-19', 'F'), ('Samuel', 'Wright', '1950-10-27', 'M'),
19 ('Chloe', 'Scott', '1948-02-12', 'F'), ('Andrew', 'Green', '1945-08-03', 'M'),
20 ('Victoria', 'Baker', '1942-11-11', 'F'), ('Christopher', 'Adams', '1940-04-09', 'M'),
21 ('Lily', 'Nelson', '1938-07-15', 'F'), ('Joshua', 'Carter', '1935-12-20', 'M'),
22 ('Zoe', 'Mitchell', '1933-06-01', 'F'), ('Nathan', 'Perez', '1930-03-10', 'M'),
23 ('Hannah', 'Roberts', '1928-09-22', 'F'), ('Ryan', 'Turner', '1927-01-05', 'M'),
24 ('Aria', 'Phillips', '1929-08-18', 'F'), ('Jack', 'Campbell', '1931-11-30', 'M'),
25 ('Layla', 'Parker', '1934-04-14', 'F'), ('Gabriel', 'Evans', '1936-07-09', 'M'),
26 ('Nora', 'Edwards', '1939-02-28', 'F'), ('Isaac', 'Collins', '1941-10-16', 'M'),
27 ('Aubrey', 'Stewart', '1943-06-12', 'F'), ('Caleb', 'Sanchez', '1946-03-01', 'M'),
28 ('Penelope', 'Morris', '1949-09-08', 'F'), ('Julian', 'Rogers', '1951-12-25', 'M'),
29 ('Stella', 'Reed', '1954-05-17', 'F'), ('Levi', 'Cook', '1956-01-20', 'M');
```

2. Include BeginAge and EndAge columns for the Rate table. The age ranges go in 5 year increments up to age 99. Separate records for male and female.

```
SQLQuery17....\praty (51))*  loq\SQLEXPR...dbo.Customer  SQLQuery16.s...Q\praty (88))*
1  -- Add age range columns
2  ALTER TABLE Rate
3  ADD
4      BeginAge INT,
5      EndAge INT;
6  GO
7
8  -- Assign age bands separately for Male and Female
9
10 WITH RateOrdered AS (
11     SELECT
12         RateID,
13         Gender,
14         ROW_NUMBER() OVER (PARTITION BY Gender ORDER BY RateID) AS rn
15     FROM Rate
16 )
17
18 UPDATE R
19 SET
20     BeginAge = (O.rn - 1) * 5,
21     EndAge = CASE
22         WHEN O.rn = 20 THEN 99
23         ELSE (O.rn * 5) - 1
24     END
25 FROM Rate R
26 JOIN RateOrdered O
27     ON R.RateID = O.RateID;
28 GO
29
30 SELECT * FROM Rate
31 ORDER BY Gender, BeginAge;
```

Results		Messages			
	RateID	Gender	Rate	BeginAge	EndAge
1	2	F	0.95	0	4
2	4	F	1	5	9
3	6	F	1.05	10	14
4	8	F	1.1	15	19
5	10	F	1.15	20	24
6	12	F	1.2	25	29
7	14	F	1.25	30	34
8	16	F	1.3	35	39
9	18	F	1.35	40	44
10	20	F	1.45	45	49
11	22	F	1.55	50	54
12	24	F	1.65	55	59
13	26	F	1.75	60	64
14	28	F	1.85	65	69
15	30	F	2	70	74
16	32	F	2.15	75	79
17	34	F	2.35	80	84
18	36	F	2.55	85	89
19	38	F	2.8	90	94
20	40	F	3.1	95	99

21	1	M	1	0	4
22	3	M	1.05	5	9
23	5	M	1.1	10	14
24	7	M	1.15	15	19
25	9	M	1.2	20	24
26	11	M	1.25	25	29
27	13	M	1.3	30	34
28	15	M	1.35	35	39
29	17	M	1.4	40	44
30	19	M	1.5	45	49
31	21	M	1.6	50	54
32	23	M	1.7	55	59
33	25	M	1.8	60	64
34	27	M	1.95	65	69
35	29	M	2.1	70	74
36	31	M	2.3	75	79
37	33	M	2.5	80	84
38	35	M	2.75	85	89
39	37	M	3	90	94
40	39	M	3.3	95	99

3. Create SQL routines that perform the following:
 - assign people an insurance rate based on birthdate (age) and gender.

```
SQLQuery21.s...Q\praty (77))*  SQLQuery20.s...Q\praty (76))*  SQLQuery18....\praty (70))*  SQLQuery17.s...Q\praty (51))*
1  --Procedure: assign insurance rate based on DOB and Gender
2
3  CREATE PROCEDURE GetCustomerInsuranceRate
4  AS
5  BEGIN
6      SET NOCOUNT ON;
7
8      SELECT
9          C.CustomerID,
10         C.FName,
11         C.LName,
12         C.Gender,
13         C.DOB,
14
15         DATEDIFF(YEAR, C.DOB, GETDATE()) -
16         CASE
17             WHEN MONTH(C.DOB) > MONTH(GETDATE())
18              OR (MONTH(C.DOB) = MONTH(GETDATE()) AND DAY(C.DOB) > DAY(GETDATE()))
19             THEN 1
20             ELSE 0
21         END AS Age,
22
23         R.Rate AS InsuranceRate
24
25     FROM Customer.dbo.Customer C
26     INNER JOIN Rate.dbo.Rate R
27     ON R.Gender = C.Gender
28     AND (
29         DATEDIFF(YEAR, C.DOB, GETDATE()) -
30         CASE
31             WHEN MONTH(C.DOB) > MONTH(GETDATE())
32              OR (MONTH(C.DOB) = MONTH(GETDATE()) AND DAY(C.DOB) > DAY(GETDATE()))
33             THEN 1
34             ELSE 0
35         END
36     ) BETWEEN R.BeginAge AND R.EndAge;
37 END;
38 GO
```

1 EXEC GetCustomerInsuranceRate;

113 %

1 0

Results Messages

	CustomerID	FName	LName	Gender	DOB	Age	InsuranceRate
1	1	Aiden	Smith	M	2026-02-10	0	1
2	2	Liam	Brown	M	2022-06-14	3	1
3	3	Noah	Johnson	M	2018-09-01	7	1.05
4	4	Emma	Davis	F	2015-11-20	10	1.05
5	5	Olivia	Miller	F	2012-04-18	14	1.05
6	6	Sophia	Wilson	F	2009-01-10	17	1.1
7	7	James	Moore	M	2006-07-22	19	1.15
8	8	Logan	Taylor	M	2003-02-14	23	1.2
9	9	Mia	Anderson	F	2000-09-30	25	1.2
10	10	Ethan	Thomas	M	1998-12-01	27	1.25
11	11	Isabella	Jackson	F	1995-05-16	30	1.25
12	12	Lucas	White	M	1992-08-09	33	1.3
13	13	Amelia	Harris	F	1989-03-03	37	1.3
14	14	Henry	Martin	M	1986-10-11	39	1.35
15	15	Charlotte	Thompson	F	1983-06-25	42	1.35
16	16	Benjamin	Garcia	M	1980-01-19	46	1.5
17	17	Evelyn	Martinez	F	1978-04-14	48	1.45
18	18	Alexander	Robinson	M	1975-09-09	50	1.6
19	19	Harper	Clark	F	1972-12-22	53	1.55
20	20	Daniel	Rodriguez	M	1970-02-07	56	1.7
21	21	Ella	Lewis	F	1968-07-30	57	1.65
22	22	Matthew	Lee	M	1965-11-18	60	1.8
23	23	Avery	Walker	F	1962-03-25	64	1.75
24	24	Joseph	Hall	M	1960-06-06	65	1.95
25	25	Scarlett	Allen	F	1958-09-14	67	1.85
26	26	David	Young	M	1955-01-08	71	2.1
27	27	Grace	King	F	1952-05-19	73	2
28	28	Samuel	Wright	M	1950-10-27	75	2.3
29	29	Chloe	Scott	F	1948-02-12	78	2.15
30	30	Andrew	Green	M	1945-08-03	80	2.5
31	31	Victoria	Baker	F	1942-11-11	83	2.35
32	32	Christop...	Adams	M	1940-04-09	86	2.75
33	33	Lily	Nelson	F	1938-07-15	87	2.55
34	34	Joshua	Carter	M	1935-12-20	90	3
35	35	Zoe	Mitchell	F	1933-06-01	92	2.8
36	36	Nathan	Perez	M	1930-03-10	96	3.3
37	37	Hannah	Roberts	F	1928-09-22	97	3.1
38	38	Ryan	Turner	M	1927-01-05	99	3.3

- Alter Customer table to include Rider flag. Compute and display rider cost - rider will increase rate by 1.5.

```

SQLQuery22.s...Q\praty (53))*  SQLQuery20.s...Q\praty (67))*  SQLQuery18....ot co
1  -- Alter Customer table to add RiderFlag
2  ALTER TABLE Customer.dbo.Customer
3  ADD RiderFlag CHAR(1);
4  GO
5
6  -- Assign RiderFlag
7  UPDATE Customer.dbo.Customer
8  SET RiderFlag =
9  CASE
10     WHEN CustomerID IN (1,2,3,4,5,6,7,8,9,10) THEN 'Y'
11     ELSE 'N'
12  END;
13  GO
14
15  -- Compute and display insurance rate with rider cost
16  SELECT
17     C.CustomerID,
18     C.FName,
19     C.LName,
20     C.Gender,
21     C.DOB,
22     C.RiderFlag,
23
24
25  -- Calculate Age
26  DATEDIFF(YEAR, C.DOB, GETDATE()) -
27  CASE
28     WHEN MONTH(C.DOB) > MONTH(GETDATE())
29     OR (MONTH(C.DOB) = MONTH(GETDATE()) AND DAY(C.DOB) > DAY(GETDATE()))
30     THEN 1
31     ELSE 0
32  END AS Age,
33
34  R.Rate AS BaseRate,
35
36  CASE
37     WHEN C.RiderFlag = 'Y' THEN R.Rate + 1.5
38     ELSE R.Rate
39  END AS TotalRate
40
41  FROM Customer.dbo.Customer C
42  INNER JOIN Rate.dbo.Rate R
43  ON C.Gender = R.Gender
44  AND (
45     DATEDIFF(YEAR, C.DOB, GETDATE()) -
46     CASE
47         WHEN MONTH(C.DOB) > MONTH(GETDATE())
48         OR (MONTH(C.DOB) = MONTH(GETDATE()) AND DAY(C.DOB) > DAY(GETDATE()))
49         THEN 1
50         ELSE 0
51     END
52  ) BETWEEN R.BeginAge AND R.EndAge;
53  GO
54  SELECT * FROM Customer.dbo.Customer

```

Results Messages									
	CustomerID	FName	LName	Gender	DOB	RiderFlag	Age	BaseRate	TotalRate
1	1	Aiden	Smith	M	2026-02-10	Y	0	1	2.5
2	2	Liam	Brown	M	2022-06-14	Y	3	1	2.5
3	3	Noah	Johnson	M	2018-09-01	Y	7	1.05	2.55
4	4	Emma	Davis	F	2015-11-20	Y	10	1.05	2.55
5	5	Olivia	Miller	F	2012-04-18	Y	14	1.05	2.55
6	6	Sophia	Wilson	F	2009-01-10	Y	17	1.1	2.6
7	7	James	Moore	M	2006-07-22	Y	19	1.15	2.65
8	8	Logan	Taylor	M	2003-02-14	Y	23	1.2	2.7
9	9	Mia	Anderson	F	2000-09-30	Y	25	1.2	2.7
10	10	Ellie	Thompson	F	1998-12-01	Y	27	1.25	2.75

	CustomerID	FName	LName	DOB	Gender	RiderFlag
1	1	Aiden	Smith	2026-02-10	M	Y
2	2	Liam	Brown	2022-06-14	M	Y
3	3	Noah	Johnson	2018-09-01	M	Y
4	4	Emma	Davis	2015-11-20	F	Y
5	5	Olivia	Miller	2012-04-18	F	Y
6	6	Sophia	Wilson	2009-01-10	F	Y
7	7	James	Moore	2006-07-22	M	Y
8	8	Logan	Taylor	2003-02-14	M	Y
9	9	Mia	Anderson	2000-09-30	F	Y
10	10	Ellie	Thompson	1998-12-01	F	Y

- Report of all customers sorted by age range, LName, FName. Displays the Rate/Rate x Rider (whichever applies - join)


```

1  -- Report of all customers sorted by Age Range, Last Name, First Name
2
3  SELECT
4      C.CustomerID, C.LName, C.FName, C.Gender, C.DOB,
5      C.RiderFlag, R.BeginAge, R.EndAge,
6
7  -- Display Rate or Rate + Rider
8      CASE
9          WHEN C.RiderFlag = 'Y' THEN R.Rate + 1.5
10         ELSE R.Rate
11     END AS FinalRate
12
13 FROM Customer.dbo.Customer C
14 INNER JOIN Rate.dbo.Rate R
15     ON C.Gender = R.Gender
16     AND (
17         DATEDIFF(YEAR, C.DOB, GETDATE()) -
18         CASE
19             WHEN MONTH(C.DOB) > MONTH(GETDATE())
20             OR (MONTH(C.DOB) = MONTH(GETDATE()) AND DAY(C.DOB) > DAY(GETDATE()))
21         THEN 1
22         ELSE 0
23     END
24     ) BETWEEN R.BeginAge AND R.EndAge
25
26 ORDER BY
27     R.BeginAge,
28     C.LName,
29     C.FName;
30 GO

```

Results		Messages							
	CustomerID	LName	FName	Gender	DOB	RiderFlag	BeginAge	EndAge	FinalRate
1	2	Brown	Liam	M	2022-06-14	Y	0	4	2.5
2	1	Smith	Aiden	M	2026-02-10	Y	0	4	2.5
3	3	Johnson	Noah	M	2018-09-01	Y	5	9	2.55
4	4	Davis	Emma	F	2015-11-20	Y	10	14	2.55
5	5	Miller	Olivia	F	2012-04-18	Y	10	14	2.55
6	7	Moore	James	M	2006-07-22	Y	15	19	2.65
7	6	Wilson	Sophia	F	2009-01-10	Y	15	19	2.6
8	8	Taylor	Logan	M	2003-02-14	Y	20	24	2.7
9	9	Anderson	Mia	F	2000-09-30	Y	25	29	2.7
10	10	Thomas	Ethan	M	1998-12-01	Y	25	29	2.75
11	11	Jackson	Isabella	F	1995-05-16	N	30	34	1.25
12	12	White	Lucas	M	1992-08-09	N	30	34	1.3
13	13	Harris	Amelia	F	1989-03-03	N	35	39	1.3
14	14	Martin	Henry	M	1986-10-11	N	35	39	1.35
15	15	Thompson	Charlotte	F	1983-06-25	N	40	44	1.35

Conclusion

Through this SQL assignment, I learned how two different databases work together. I got a better understanding of how databases are structured and why organization day properly matters. I also learned how different tables can connect with each other and how important it is to make sure the right data matches when working across multiple tables or databases. This assignment also showed me how small mistakes, like wrong names or syntax, can cause errors, so being careful and checking details matters a lot. Overall, I now understand SQL better as a tool for managing, connecting, and updating data in an organized way.